Student Performance Analytics System

Project Documentation

Prepared according to the EWB Courses Python with AI project brief

1. Project Title and Student Details

Project Title: Student Performance Analytics System

Student Name: Krishna Kumar Dubey

Course / Department: B.Tech CSE

Institution: Jagannath University,Jaipur

Project Category: Python data-analysis mini project

2. Objective

The objective of this project is to analyse student performance using Python. The program reads student information and marks from a CSV dataset, calculates totals and averages, assigns grades, determines pass/fail status, summarizes class and subject-wise performance and attendance, and exports the processed results to a new CSV file.

3. Technologies Used

- Python: functions, conditional statements, loops, and program flow.
- Pandas: loading CSV data into a DataFrame, calculating columns, applying functions, selecting top records, and exporting reports.
- NumPy: numerical summaries such as mean, minimum, and maximum.
- CSV: input dataset and output report.
- Visual Studio Code: editing and running the Python program.

4. Dataset Description

The input dataset is students.csv. It contains student ID, name, department, marks in five subjects, and attendance percentage. Expected columns are Student_ID, Name, Department, Python, SQL, Java, CSS, JavaScript, and Attendance. Marks are expected to be numeric values out of 100, and attendance is a percentage from 0 to 100.

Column	Description
Student_ID	Unique identifier for each student
Name	Student name
Department	Department, e.g. CSE or IT
Python	Python marks out of 100
SQL	SQL marks out of 100
Java	Java marks out of 100
CSS	CSS marks out of 100
JavaScript	JavaScript marks out of 100
Attendance	Attendance percentage from 0 to 100

Dataset note: the sample records prepared for this project are synthetic data, not verified real college records. State this clearly in the final submission if synthetic records are used.

The program can process additional rows without code changes when the required columns remain consistent and data values are valid.

5. Implementation

Import libraries

Pandas is imported as pd for tabular data handling, and NumPy is imported as np for numerical calculations.

Define subjects

The SUBJECTS list contains Python, SQL, Java, CSS, and JavaScript so the same subject names are reused throughout the program.

Load dataset

load_data() reads students.csv using pd.read_csv() and returns a Pandas DataFrame.

Calculate marks

calculate_marks(df) sums the five subject marks row-wise to create Total and calculates their mean to create Average, rounded to two decimal places.

Assign grades

assign_grade(average) applies project-defined grade bands: O at 90 or above, A+ at 80–below 90, A at 70–below 80, B+ at 60–below 70, B at 50–below 60, C at 40–below 50, and F below 40.

Check pass/fail

check_result(row) returns Fail if any of the five subject marks is below 35; otherwise, it returns Pass.

Analyse data

analyze_data(df) adds Grade and Result columns, prints the student report, calculates class statistics, pass/fail counts, subject averages, top three students, and average attendance.

Save report

main() calls the functions in order and writes the processed DataFrame to student_performance_report.csv with index=False.

6. Key Features

- Automatic total and average calculation for five subjects.
- Grade assignment based on average marks.
- Pass/fail checking based on subject-wise passing marks.
- Class average, highest average, and lowest average.
- Pass and fail student counts.
- Subject-wise average marks.
- Top three students ranked by average marks.
- Average attendance calculation.
- Export of processed data to a separate CSV report.

7. Output / Screenshots

The project was run successfully in the user's environment. The program prints a Student Performance Report, Class Statistics, Pass/Fail Report, Subject-wise Average, Top 3 Students, and Attendance Report in the terminal. It also creates student_performance_report.csv.

Add your own screenshots before submission: (1) program running in the VS Code terminal, (2) terminal reports and statistics, and (3) student_performance_report.csv opened in the editor.

Exact numerical results are intentionally not listed here; copy them from the actual output generated by your dataset.

8. Final Outcome

The project automates basic student-mark calculations and creates a structured performance report. It demonstrates how Python, Pandas, and NumPy can be used to read a dataset, transform records, calculate statistics, and save analysis results.

9. Challenges and Learning

- Keeping the CSV column names consistent with the Python code.
- Calculating totals and averages row-wise using Pandas.
- Organising the program with reusable functions.
- Using conditional logic to assign grades and determine pass/fail status.
- Using NumPy for class-level numerical summaries.
- Saving processed records to a new CSV file without the DataFrame index.

Add any other challenges you personally encountered and how you solved them. Do not include experiences that did not occur.

10. GitHub Repository

Suggested repository name: student-performance-analytics

Repository URL: https://github.com/Krishna331457/Student-Performance-Analysis

Upload main.py, students.csv, README.md, and this documentation. Verify that the project runs and the files are visible, then submit the repository URL through the Google Form provided by EWB Courses.

Project Rules and Limitations

The grade bands and 40-mark subject passing threshold are rules defined for this project and may differ from an institution's official policy. The input file must contain the required columns and numeric marks. The current version creates text output and a CSV report; it does not implement machine learning or visual charts.